

PV102 Day 2

Site and Solar Resource Assessment

Measure the site before you size the system.

CTS Day 18 | Tuesday, August 4, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

• Students convert vague site assumptions into observable site conditions: solar access, shade, orientation, tilt, space, weather exposure, access, roof or ground constraints, and maintenance realities.

- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.

Do not guess when an unknown can be recorded.



Measure the site before you size the system.

Learning targets

By 3:00 PM students should be able to:

- Identify site factors that affect PV production and installation practicality.
- Use orientation, tilt, shade, and access observations to support a design decision.
- Separate measured observations from assumptions and preferences.
- Create a site-note handoff that Day 3 can use for load and energy work.



Evidence beats confidence. A tragic lesson for civilization, apparently.

Measure the site before you size the system.

Morning readiness check

Quiz first, then review as evidence.

- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.



The quiz measures instruction as much as students.

Core rule

- Measure the site before you size the system.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.

One rule, repeated until reality behaves.

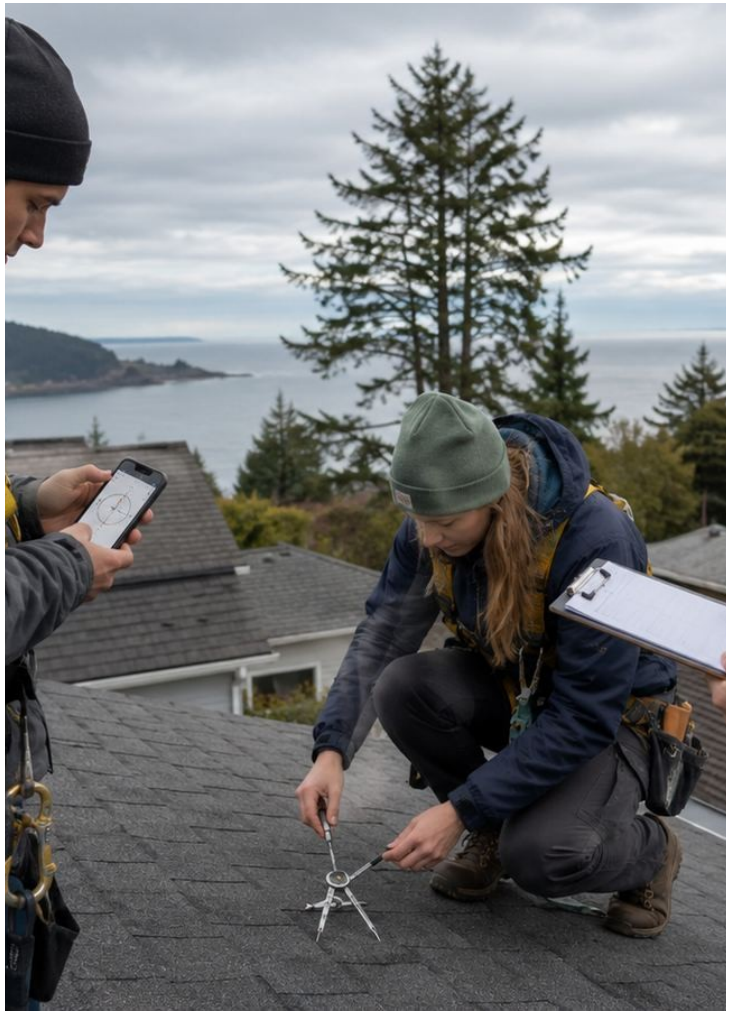


Measure the site before you size the system.

What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.



Measure the site before you size the system.

New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



Measure the site before you size the system.

Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

Measure the site before you size the system.

Decision checks

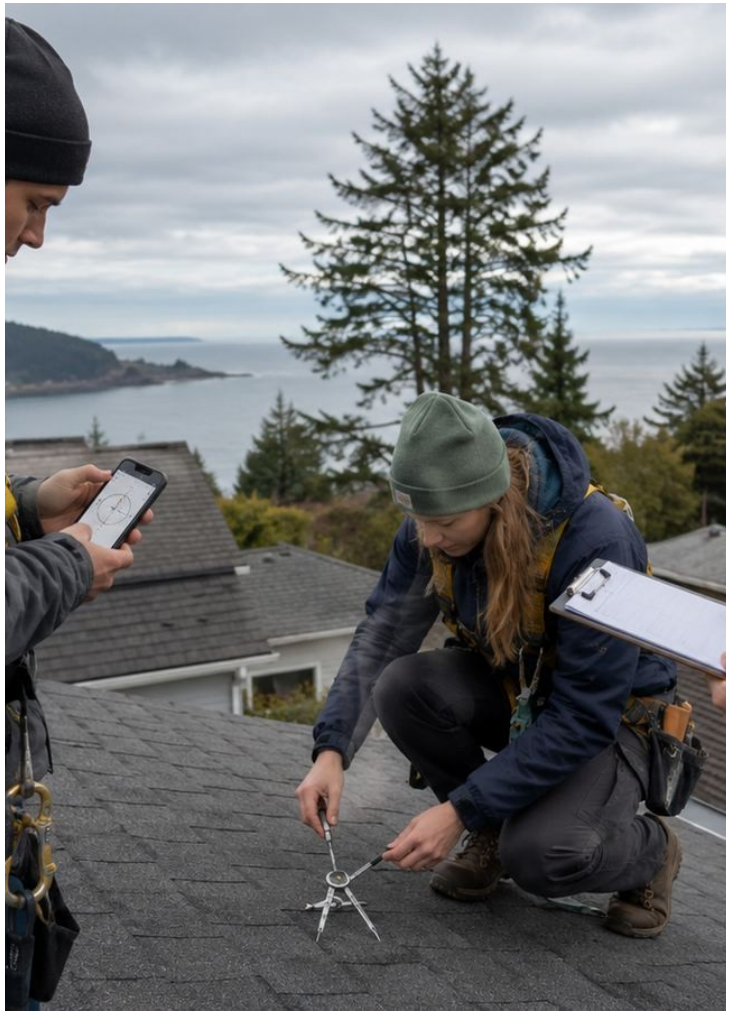
Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

Measure the site before you size the system.

Guided example

Model one complete thinking path.

- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.



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Team roles

Three teams of two, roles rotate after the first pass.

- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.



Measure the site before you size the system.

Lab mission

Site Walk and Solar Access Handoff

- Teams inspect three candidate locations and decide which one deserves further design work.



This is the work block. Keep it practical and evidence-based.

Measure the site before you size the system.

Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- South-facing roof plane with one morning tree shadow and limited attic access.
- Open ground rack area with excellent sun but long wire run and mower traffic.
- West-facing shed roof near the load with easy service access and afternoon sun.

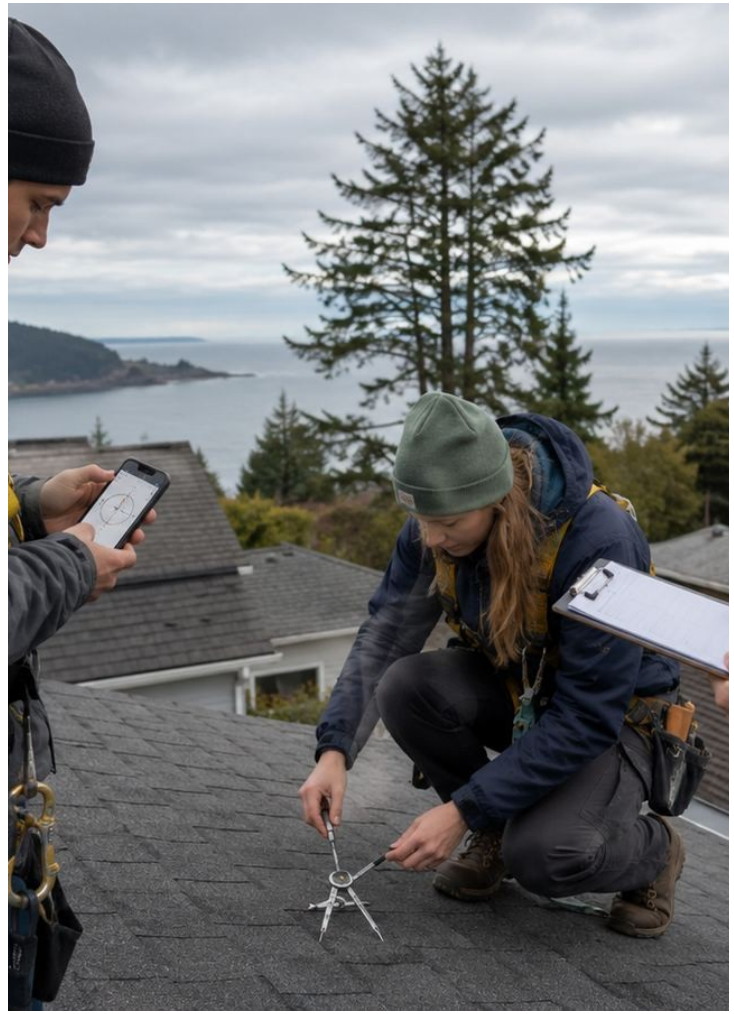


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Lab procedure

Run the same loop every day.

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.



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Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.



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Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.

Stop, isolate, verify, ask. The glamorous life.

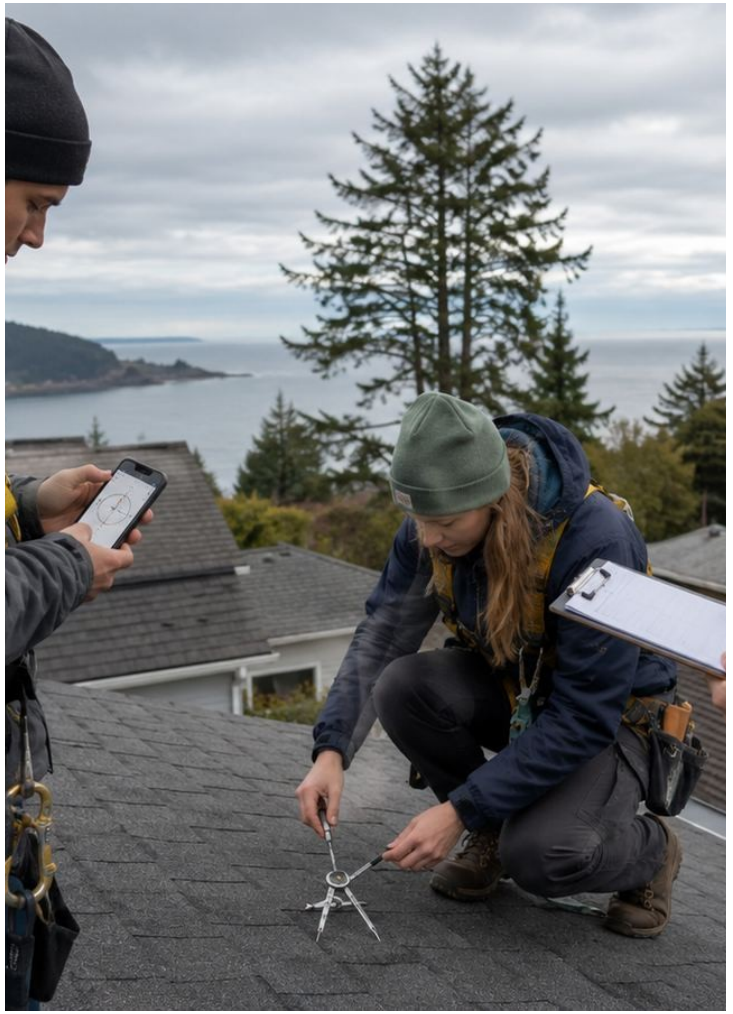


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Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?



Measure the site before you size the system.

Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?



Measure the site before you size the system.

Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.

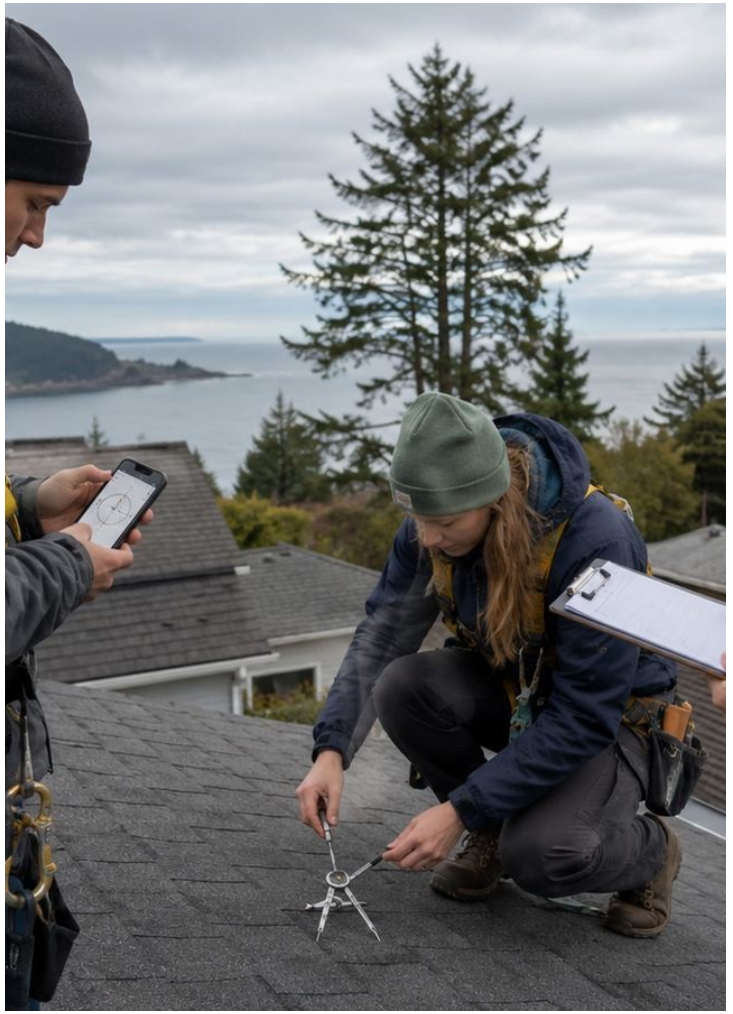


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Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.

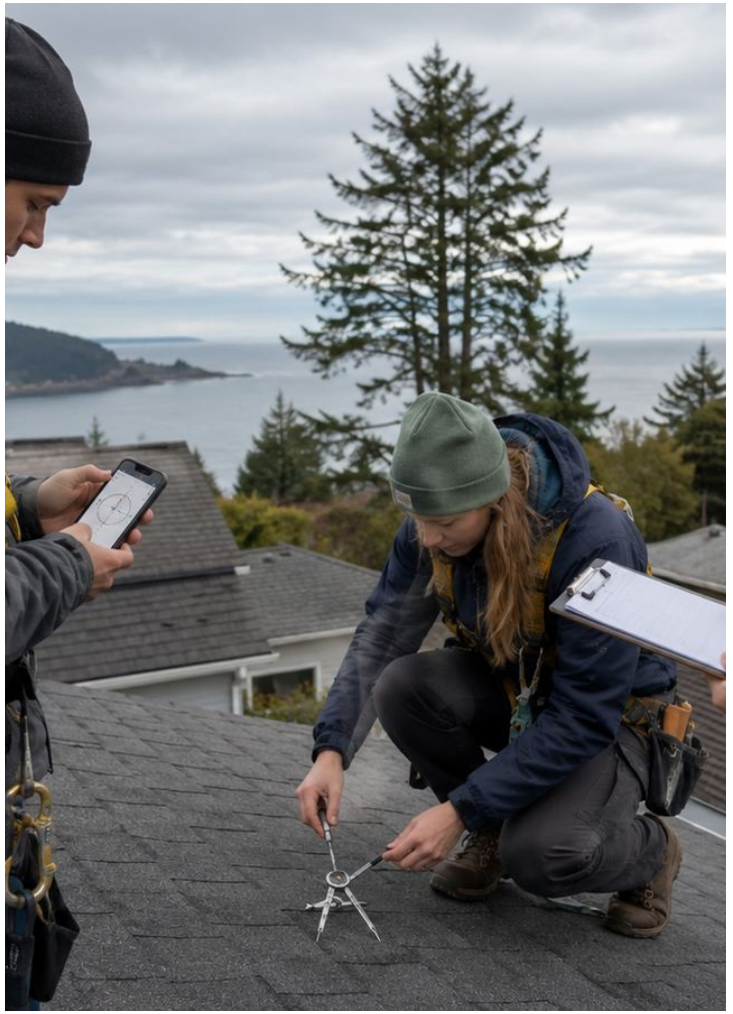


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Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.

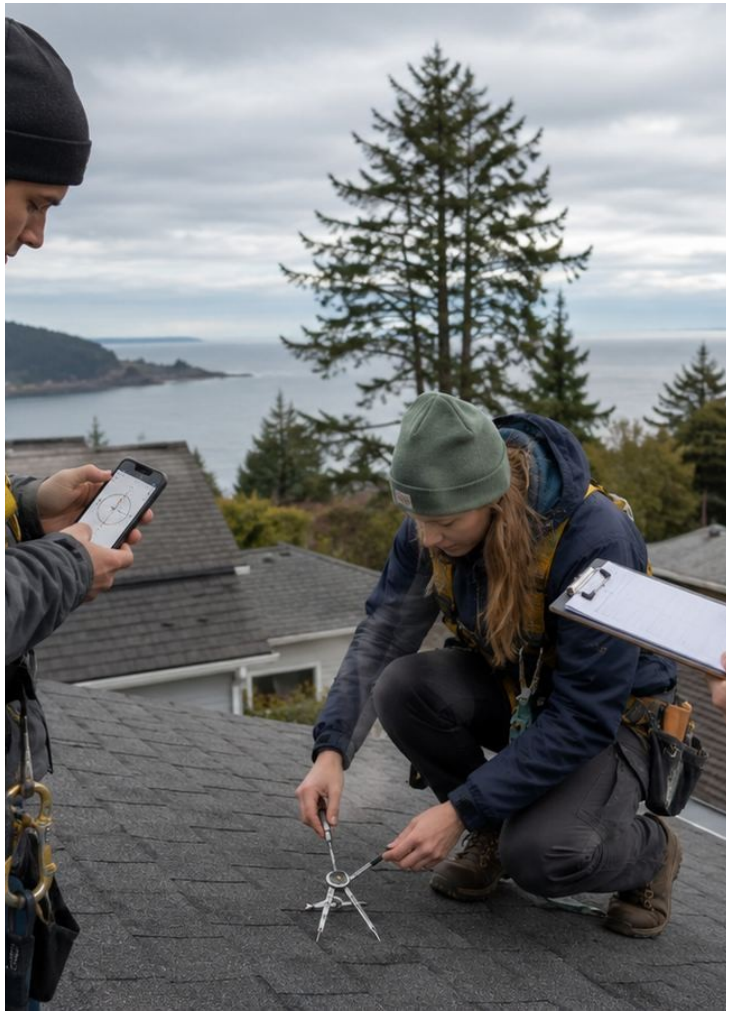


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Exit ticket

Individual close-out

- Which site condition would most likely force a redesign, and what evidence supports that answer?



Collect before reset so the day leaves a paper trail. Weirdly useful.

Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 Day 3 uses the site handoff to build a load profile and first-pass energy budget.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.



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Final reminder

What professionals do

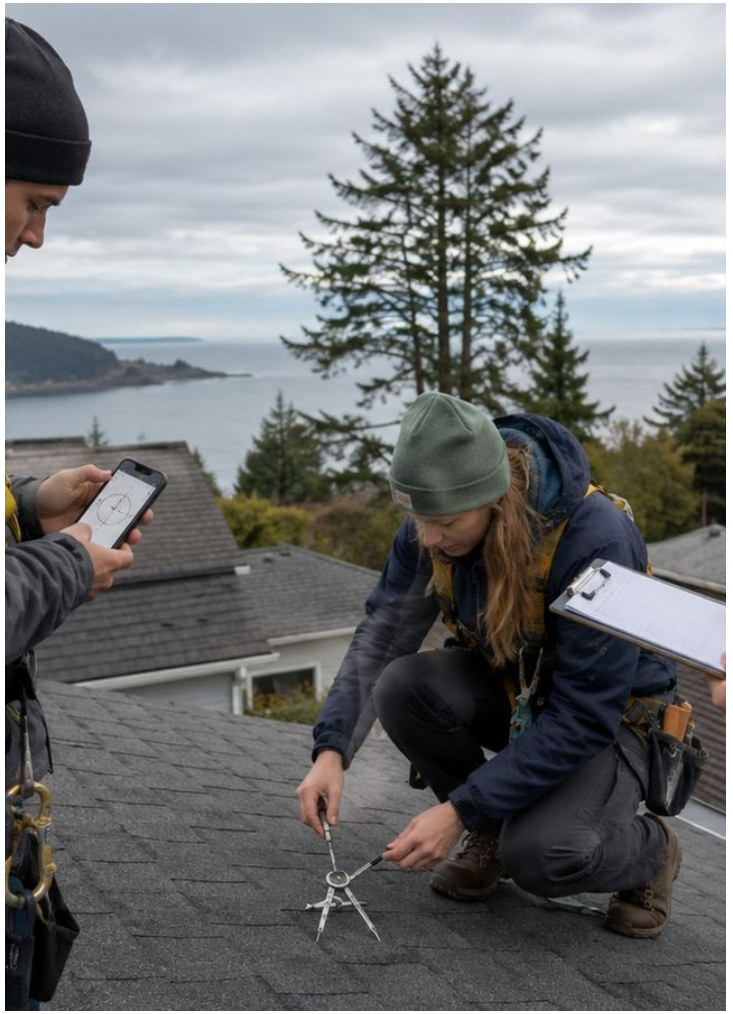
- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.



Measure the site before you size the system.

Close

- Measure the site before you size the system.



Reset tools. Collect packets. Archive the handoffs.

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